

Hybrid Quantum–Classical Pauli-Correlator Architecture for Image Reconstruction

Sparsho Chakraborty¹ and Siddhartha Patra^{2*}

¹School of Basic Sciences, Department of Physics, Indian Institute of Technology Bhubaneswar, Odisha, India.

^{2*}Centre for Quantum Engineering, Research and Education, TCG CREST, Kolkata, India.

*Corresponding author(s). E-mail(s): siddhartha.patra@tcgcrest.org;
Contributing authors: 25ph05023@iitbbs.ac.in;

Abstract

We introduce the Pauli-Correlator Autoencoder (PCAE), a hybrid quantum–classical architecture for image reconstruction. PCAE assembles matrix-product-state patch features, a variational quantum circuit exposing a Pauli-correlator latent space, two-dimensional Fourier positional encoding, a learnable Hamiltonian energy diagnostic, and a classical decoder into a single pipeline. Two instantiations, PCAE1D and PCAE2D, realise chain and grid interaction graphs. Group-averaged observable pooling makes the PCAE2D feature map exactly equivariant to the eight symmetries of the square. The contribution is this specific assembly rather than any ingredient in isolation. Across six real image datasets, per-image-trained PCAE2D models reconstruct with peak signal-to-noise ratio (PSNR) between 25.8 and 41.6 decibels. On a resource-matched, multi-seed held-out comparison using CIFAR-10, an equivalence procedure using two one-sided tests (TOST), with Holm correction for seven comparisons, establishes that PCAE2D is statistically competitive with four resource-matched controls. Five fitted checkpoints are replayed on superconducting quantum hardware without retraining, reaching 25.90 to 31.52 decibels. PCAE adds a pair-observable channel that is necessary under a fixed linear readout on a leakage-audited task requiring distant-patch products, an interpretable Hamiltonian energy diagnostic, and native hardware execution. The results establish PCAE as competitive with classical baselines at the six-qubit scale.

Keywords: Quantum image processing, Tensor networks, Variational quantum algorithms, Pauli observables, Group-equivariant representations, Quantum hardware pilot

[Rename: the model was the Hamiltonian Vision Kernel (HVK). HVK is not a kernel method, so it is now the Pauli-Correlator Autoencoder (PCAE), with variants PCAE1D and PCAE2D. Only the first uses are marked. All other uses of HVK were replaced directly. Student: (1) regenerate every figure whose labels show HVK, HVK1D or HVK2D, (2) rename the model in the supplement, including its title, (3) then update the three places where this paper cites the supplement’s title.]

Contents

1	Introduction	2
2	Pauli-Correlator Autoencoder: Architecture	5
3	Experimental Setup	8
4	Results	8
4.1	Reconstruction across six real image datasets	8
4.2	Scope of per-image fitting	9
4.3	Competitive with resource-matched classical baselines	9
4.4	A real hardware reconstruction pilot	11
4.5	Noise/shot trade-off: a free simulator complement	13
4.6	Real-hardware anchor points: repeated execution and a second backend	14
4.7	Exact D_4 -equivariant observable pooling	15
4.8	Pair-observable necessity in the nonlocal-structure protocol	17
4.9	A scoped training-dynamics diagnostic	18
5	Discussion	18
5.1	Effect of the Hamiltonian energy term	18
5.2	Task-dependent representational scope	19
5.3	Topology alignment as an inductive bias	20
5.4	Validated scope	21
6	Conclusion	22

1 Introduction

Hybrid quantum–classical models are now often proposed for vision tasks. The aim is to obtain representations with a favourable inductive bias. Such models can use entanglement, physically motivated regularisers, or tensor-network structure, which standard convolutional networks do not have (Cerezo et al. 2021; Tilly et al. 2022; Preskill 2018). The variational quantum eigensolver and its later variants use the fact that physical Hamiltonians split into local terms. This allows them to build shallow circuits suited to the problem (Cerezo et al. 2021; Tilly et al. 2022; Bauer et al. 2020). Quantum image processing gives several ways to store pixel values directly in qubits. These range from amplitude-style encodings (Le et al. 2011) to basis-encoded

pixel values (Zhang et al. 2013). Basic image operations on these encodings have also been developed (Chakraborty et al. 2022; Zhou et al. 2017; Yao et al. 2017). Tensor-network methods work well as compressed representations of structured classical data, such as images (Cirac et al. 2021; Latorre 2005; Han et al. 2018; Stoudenmire and Schwab 2016; Ran et al. 2020; Guala et al. 2023) and particle-physics events (Araz and Spannowsky 2022). Hybrid models have also been applied to related imaging tasks, such as tomographic reconstruction (Haga 2024; Nau et al. 2023) and ghost imaging (Zhai et al. 2025). Quantum convolutional and capsule networks are further examples (Slabbert and Petruccione 2024; Cong et al. 2019; Hur et al. 2022; Liu et al. 2023). PCAE focuses on the combination of a Pauli-correlator latent space, enforced group-equivariant pooling, and a physically motivated energy diagnostic.

We introduce the Pauli-Correlator Autoencoder (PCAE1D/PCAE2D), which combines these ingredients into a single architecture with three properties studied jointly. The first is a *Pauli-correlator latent space*, in which measured VQC observables rather than a raw quantum state form the representation handed to a classical decoder, following quantum-enhanced feature maps (Havlíček et al. 2019). The second is an *exact, numerically verified D_4 -equivariant observable-pooling construction*, which group-averages the observable grid over the eight symmetries of the 4×4 patch grid rather than motivating symmetry through architecture choices alone. This follows group-equivariant classical vision (Cohen and Welling 2016; Bronstein et al. 2021) and the equivariant-quantum-neural-network line that formalises symmetry-embedding constructions for variational circuits (Meyer et al. 2023; Nguyen et al. 2024; Schatzki et al. 2024; West et al. 2024, 2025). The third is a *learnable Hamiltonian energy diagnostic*, with full nearest-neighbour $XX/YY/ZZ$ sectors in PCAE1D and grid-edge ZZ terms in PCAE2D. Table 10 contrasts these properties against classical, adversarial, and standard quantum autoencoder baselines.

Novelty relative to the nearest hybrid-vision architectures

PCAE’s closest architectural relatives are three hybrid quantum-vision patterns already established individually in the literature.

Register-based quantum autoencoders (Romero et al. 2017; Wang et al. 2024; Asaoka and Kudo 2025) encode an image’s pixel amplitudes directly into a qubit register and train a trash-qubit fidelity objective. PCAE feeds the circuit only quantum-inspired, pre-compressed MPS patch features, not raw pixel amplitudes. It hands the decoder measured Pauli correlators rather than register amplitudes or a fidelity score. PCAE is an autoencoder in the usual encoder–decoder sense. It maps each image patch to a measured observable vector and decodes it back to pixels. It does not compress a quantum state.

Quantum-kernel vision models (Havlíček et al. 2019; Beaulieu et al. 2022) embed an input into a quantum feature space and read out a scalar kernel value consumed by a classical kernel method. PCAE reads out an explicit vector of local and pairwise Pauli expectation values as a latent representation for a trained decoder, and adds enforced group-equivariant pooling and a Hamiltonian energy diagnostic that a kernel construction has no analogue for.

Tensor-network (TN) circuit classifiers (Guala et al. 2023; Stoudenmire and Schwab 2016; Han et al. 2018) use MPS structure either as a standalone classical model or as a template for a shallow quantum circuit trained for classification. PCAE uses MPS compression only as a quantum-inspired *preprocessing* step that sets a separately trained VQC’s rotation angles, and targets per-image reconstruction rather than classification.

PCAE’s contribution is the assembly of MPS preprocessing and the three properties above in one pipeline (Table 10). Section 5.2 examines how the individual ingredients recur across this literature.

Contributions

1. **Architecture.** PCAE1D/PCAE2D combines MPS patch features, a six-qubit VQC, a 19–27-dimensional Pauli-correlator vector, 2D Fourier positional encoding, a graph-Hamiltonian diagnostic, and a classical decoder.
2. **Reconstruction scope (per-image fitting).** Identically configured per-image fitting experiments cover six real image datasets (Section 4.1).
3. **Competitive with resource-matched classical baselines (held-out comparison).** An equivalence procedure using two one-sided tests (TOST), with Holm correction for seven comparisons, shows that PCAE2D meets the ± 1 dB equivalence criterion against four of seven distinct resource-matched classical/ablated controls and substantially outperforms the two null controls (strict classical random features and a random-VQC) (Section 4.3).
4. **Hardware pilot.** Five fitted checkpoints are replayed on IBM hardware without retraining and decoded from hardware-measured observables (Section 4.4), strengthened by a noise/shot-budget sweep and repeated execution across backends (Sections 4.5–4.6).
5. **Equivariant observable pooling (proposition).** An exact, numerically verified D_4 -equivariant observable-pooling construction makes the PCAE2D observable map equivariant to the eight symmetries of the square, to floating-point precision (Section 4.7). The guarantee follows from Reynolds group averaging over the D_4 orbit rather than from anything quantum, and the same construction gives the same exactness for a purely classical feature map (Supplementary Material). We report it as a design property of the pipeline.
6. **Restricted nonlocal result (diagnostic).** In a leakage-audited protocol with a fixed linear readout, pair observables are necessary when the target depends explicitly on distant-patch products (Section 4.8).

A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, reports resource-matched held-out comparisons of PCAE and classical/ablated feature maps, together with full hardware-pilot methodology covering circuit construction, validation, job identifiers and quota accounting. It also gives provenance audits of excluded exploratory results, among them the training-dynamics traces of Section 4.9, which we report as an interpretability readout rather than as evidence of a transition. Section 4.3

summarises the central finding of the held-out comparison and how it relates to the results here.

2 Pauli-Correlator Autoencoder: Architecture

PCAE treats each spatial patch as a finite spin state and compresses it as a matrix product state (MPS). A variational quantum circuit (VQC) then measures a fixed set of Pauli observables, which a classical multi-layer perceptron (MLP) decodes back into pixels. A learnable graph-Hamiltonian energy term regularises training. Parameterised Hamiltonian learning approaches treat the Hamiltonian as the primary learned object (Shi et al. 2023); in PCAE it is instead a diagnostic regulariser alongside a separate reconstruction objective.

Problem statement.

Given an image x partitioned into non-overlapping patches $\{x_p\}_{p=1}^P$, PCAE learns a per-patch reconstruction map. Each patch is compressed to a quantum-inspired MPS feature vector $\phi_{\text{MPS}}(x_p)$, a learned linear layer sets the VQC rotation angles, and measuring a fixed set of Pauli observables yields the correlator feature map $\Phi(x_p) \in \mathbb{R}^d$ ($d = 27$ for PCAE1D, 19 for PCAE2D). With $n = 6$ qubits, bond set $\mathcal{E}$, and c measured pair channels per bond,

$$d = 2n + c|\mathcal{E}| = \begin{cases} 2(6) + 3(5) = 27, & \text{PCAE1D (ZZ, XX, YY on a 5-bond chain),} \\ 2(6) + 1(7) = 19, & \text{PCAE2D (ZZ on 7 grid edges),} \\ 2(6) + 1(5) = 17, & \text{ZZ-only ablation control.} \end{cases} \quad (1)$$

All reported PCAE1D and PCAE2D results use $d = 27$ and $d = 19$. A classical decoder D_θ maps $\Phi(x_p)$, together with a Fourier positional encoding $\pi(p)$, back to a reconstructed patch $\hat{x}_p = D_\theta(\Phi(x_p), \pi(p))$. Training minimises the pixel-reconstruction loss with the graph-Hamiltonian energy term $E(\cdot)$ as an auxiliary regulariser,

$$\mathcal{L}(\theta) = \frac{1}{P} \sum_{p=1}^P \|x_p - \hat{x}_p\|_2^2 + \lambda \frac{1}{P} \sum_{p=1}^P E(\Phi(x_p)), \quad (2)$$

where the second term is the mean graph-Hamiltonian energy over patches and λ sets its weight (Section 5.1 reports its effect). The shared pipeline is

$$\begin{aligned} \text{Image} &\rightarrow \{\text{Patches}\} \rightarrow \text{MPS} \\ &\rightarrow 46\text{-D}/22\text{-D Feature} \rightarrow 6\text{-D Vector} \\ &\rightarrow 27\text{-D}/19\text{-D Latent} \rightarrow \text{Decoder} \\ &\rightarrow \text{Reconstruction.} \end{aligned} \quad (3)$$

Why MPS features.

Treated as an amplitude-encoded quantum state, a patch of n sites has a coefficient tensor with 2^n entries. Matrix-product-state factorisation instead represents it as a

Table 1 The two executed configurations. Both use six qubits and the same decoder; they differ in patch size and in the measured observable set.

	PCAE1D	PCAE2D
Benchmark	<i>Monalisa</i>	multi-dataset
Image size	256×256	32×32 (native)
Patch size	64×64	8×8
MPS sites	12	6
Local X/Z expectations	24	12
Nearest-neighbour ZZ	11	5
Bipartite entropies	11	5
MPS feature vector	46-D	22-D
Measured observables	27-D	19-D
local Z , local X	6, 6	6, 6
pair terms	5 each ZZ, XX, YY	7 grid-edge ZZ
Fourier positional vector	8-D	4-D

chain of small tensors bounded by a fixed bond dimension χ . This is a compact, singular-value-controlled compression whose entanglement-entropy profile is already known to track image compressibility (Latorre 2005; Stoudenmire and Schwab 2016), and whose local expectations and correlations are cheap features that do not require a full state-preparation circuit.

Why a Pauli-correlator latent space rather than a raw quantum state.

Handing the classical decoder a fixed vector of measured Pauli expectation values, instead of full state tomography, keeps the quantum-to-classical interface at a size linear in the qubit count, on the bounded-degree graphs used here, rather than exponential readout cost. The execution cost also grows with the number of measurement settings and the shots per setting. It also follows the measured-observable-as-feature pattern used by quantum-enhanced feature maps generally (Havlíček et al. 2019). This choice is what makes the interpretability diagnostics of Section 5.1 and the training-dynamics diagnostics of the companion Supplementary Material possible, since a full state vector would not expose a comparably compact, physically meaningful summary.

Why a grid rather than a chain interaction topology for PCAE2D.

Aligning the six-qubit interaction graph with the 2×3 patch-neighbour adjacency, rather than an arbitrary linear chain, gives the latent a two-dimensional coupling pattern. The D_4 pooling of Section 4.7 does not depend on this choice. It acts by permuting the 4×4 grid of patches in an image, and a 2×3 graph has no 90° rotation symmetry of its own. PCAE1D’s chain topology is the control without a grid. It is used to measure what the grid adds (Section 5.3).

Two preprocessing configurations were executed, one per topology. Both follow the same three stages. Each image is first divided into sixteen non-overlapping patches.

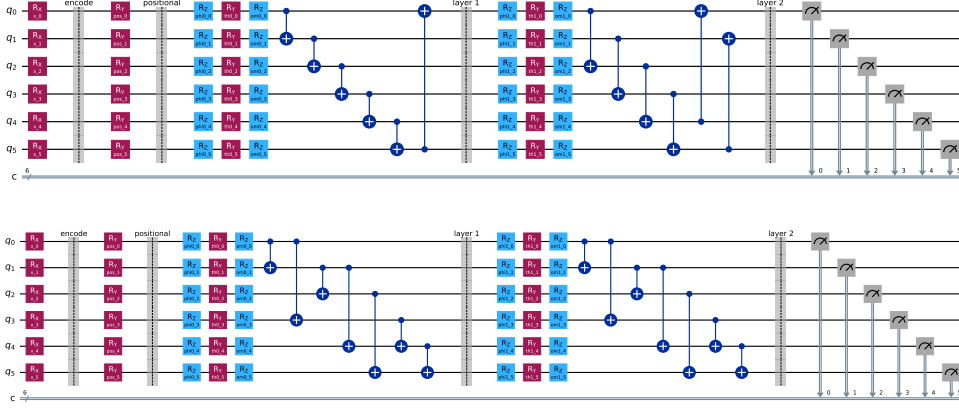


Fig. 1 The PCAE1D (top) and PCAE2D (bottom) variational ansätze, exactly as executed (encoding R_X rotations, positional R_Y rotations, then two entangling layers of R_Z - R_Y - R_Z single-qubit rotations with a CNOT ring for PCAE1D or fixed grid-edge CNOTs for PCAE2D). Both circuits were rebuilt gate-for-gate in Qiskit and validated against the original PennyLane training circuit before use in the hardware pilot of Section 4.4.

Every patch is then ℓ_2 -normalised and sequentially SVD-compressed to a matrix product state of bond dimension $\chi = 4$. Finally, the local expectations, nearest-neighbour correlations and bipartite entropies of that state are read off as the quantum-inspired MPS feature vector. The two configurations differ only in the sizes involved, which Table 1 lists in full. The 2^n pixel values of a patch ($n = 6$ for 8×8 , $n = 12$ for 64×64) are flattened in row-major order and reshaped into an order- n tensor with binary indices, so each index is one site of dimension 2. Sequential SVDs from the left leave the MPS in left-canonical form, with the orthogonality centre on the last site, and all features are read off in this gauge. **[Student: all-zero patches do occur (MNIST corners are black). The code avoids them by adding 10^{-4} to every pixel before the MPS step (run_full_dataset_sameset.py: patches + 1e-4). Please state this offset here. Please also confirm the canonical form against quimb’s compress default.]** The remaining stages are common to both. A learned linear layer compresses the MPS feature vector to six circuit angles, one per qubit. After the circuit is executed, a two-hidden-layer MLP (128 and 256 units) maps the measured observables together with the Fourier positional vector back to a single greyscale patch. The positional encoding follows established use as a spatial inductive bias in classical generative vision models (Xu et al. 2021). Fourier features of this kind also help coordinate-based networks fit fine image detail (Tancik et al. 2020). During training the circuit is simulated exactly, so the decoder receives exact expectation values. Finite-shot estimates enter only in the hardware replay and the noisy simulation of Sections 4.4–4.6. Figure 1 shows the exact PCAE1D and PCAE2D ansatz circuits. These were rebuilt in Qiskit for the hardware pilot of Section 4.4, and numerically verified against the training-time PennyLane circuit before use. The complete Hamiltonian, loss, positional-encoding and measurement definitions are retained in the companion supplement.

Table 2 PCAE variants tested in this paper.

Variant	Topology	Latent dim.	Symmetry
PCAE1D	6-qubit chain	27-D	None (control)
PCAE2D	2×3 grid	19-D	D_4 -motivated only
Symmetric PCAE1D	6-qubit chain	27-D	U(1) coupling
D_4 -pooled PCAE2D	2×3 grid, pooled	19-D	D_4 -equivariant (exact)

Table 2 summarises the tested variants. PCAE1D uses a linear 6-qubit chain (5 nearest-neighbour bonds, 27-D observable vector); PCAE2D arranges the same 6 qubits on a 2×3 lattice matching patch-boundary adjacency (19-D latent signature); a symmetric PCAE1D variant adds a U(1)-preserving coupling constraint; and a D_4 -pooled PCAE2D variant (Section 4.7) group-averages the observable grid over the eight symmetries of the 4×4 patch grid, $\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx)$, which is equivariant by construction. The IBM Cloud circuit probe reported in the supplement confirms both PCAE1D and PCAE2D compile to depth-18, 6-qubit circuits on a Heron-class basis.

3 Experimental Setup

The reconstruction results in Sections 4.1, 4.2 and 4.4–4.6 use the single-image *Monalisa* protocol developed for PCAE. A fresh model is optimised directly on its target image, with no held-out split. These results therefore measure per-image fitting and reconstruction capability. The hardware pilot replays the resulting fitted checkpoints on real devices without retraining. The symmetry and restricted pair-observable diagnostics use the separate protocols stated in their respective sections. The training-dynamics readouts of Section 4.9 are reported as an interpretability diagnostic, with their scope set by the negative control given there. Held-out generalisation and component necessity are evaluated under the resource-matched statistical protocol of the companion supplementary study (Section 4.3). All settings report MSE and PSNR = $20 \log_{10}(1/\sqrt{\text{MSE}})$; image settings additionally report SSIM. Full implementation settings and environment versions are given in the companion supplementary document.

4 Results

4.1 Reconstruction across six real image datasets

For each image, we trained a fresh PCAE2D model using non-overlapping 8×8 patches (16 patches/image, 80–100 optimisation steps). The sample covers CIFAR-10 (Krizhevsky 2009), MNIST (LeCun et al. 1998), Fashion-MNIST (Xiao et al. 2017), and the PathMNIST, BloodMNIST, and PneumoniaMNIST subsets of MedMNIST v2 (Yang et al. 2023). Table 3 reports mean PSNR between 25.8 and 41.6 dB (SSIM 0.91–1.00). This establishes that the architecture fits varied image structures under a

Table 3 PCAE2D same-set reconstruction across six real image datasets (per-image trained, 80–100 steps, mean $\pm$ std over sampled images).

Dataset	n images	PSNR (dB)	SSIM
CIFAR-10	3	37.75 ± 2.38	0.9986 ± 0.0003
MNIST	3	25.77 ± 2.23	0.9783 ± 0.0116
Fashion-MNIST	3	34.55 ± 3.05	0.9980 ± 0.0006
PathMNIST	3	36.86 ± 2.11	0.9118 ± 0.0627
BloodMNIST	2	36.57 ± 2.36	0.9922 ± 0.0063
PneumoniaMNIST	2	41.60 ± 0.98	0.9985 ± 0.0006

common protocol. This protocol is per-image fitting, in the same sense as implicit neural representations, which fit one network to one image or scene (Ulyanov et al. 2020; Sitzmann et al. 2020; Tancik et al. 2020; Mildenhall et al. 2020; Dupont et al. 2021; Strümppler et al. 2022). Quantum versions of this idea fit a single signal or image with a variational circuit (Zhao et al. 2024; Fujihashi and Koike-Akino 2024; Lizzio Bosco et al. 2026). The closest of these, QIREN, is a quantum coordinate network aimed at a fitting advantage over classical Fourier networks (Zhao et al. 2024). PCAE is instead a patch autoencoder with a measured-correlator bottleneck, and it aims at competitiveness with classical baselines. These results therefore measure representation capacity. Generalisation to held-out images is tested in Section 4.3. A compression claim would also need a bit rate, which we do not report (Dupont et al. 2021; Strümppler et al. 2022).

4.2 Scope of per-image fitting

A model optimised on one image is specific to that image. On *Monalisa*, zero-shot transfer to a second image reaches 8.31 dB PSNR, with SSIM 0.1044. Extending training to include the second image recovers 28.63 dB, with SSIM 0.9858. The per-image results above therefore measure per-image optimisation and reconstruction capability. Section 4.3 tests the held-out case directly.

4.3 Competitive with resource-matched classical baselines

The results above measure per-image fitting. A separate question is whether PCAE’s feature map improves reconstruction on real, *held-out* images under a resource-matched multi-seed protocol. The full study is in the companion document, *Supplementary Material for the Pauli-Correlator Autoencoder: Resource-Matched Ablations, Validation, and Reproducibility*. Its main results are summarised below, including the equivalence tests of Table 4.

On held-out CIFAR-10, the raw/local linear control reaches 18.80 ± 1.42 dB, against PCAE2D’s 18.12 ± 1.54 dB. Aggregating held-out images within each split gives a PCAE2D-minus-control difference of -0.68 dB, with bootstrap 95% CI

Table 4 TOST equivalence test on held-out CIFAR-10 (± 1 dB margin, $\alpha = 0.05$, five seed-level paired differences, PCAE2D minus control). Equivalence requires the 90% CI to lie inside $[-1, +1]$ dB. Values from the Supplementary Material.

Control	Δ PSNR (dB)	90% CI (dB)	TOST p	Holm p	Verdict
Raw/local linear	-0.68	$[-0.96, -0.40]$	0.034	0.103	Not equivalent
No entanglement	-0.34	$[-0.69, +0.02]$	0.008	0.032	Equivalent
ZZ-only observables	-0.44	$[-0.61, -0.27]$	0.001	0.007	Equivalent
Quadratic classical	-0.21	$[-0.46, +0.03]$	0.001	0.007	Equivalent
Shuffled pair observables	-0.19	$[-0.35, -0.02]$	< 0.001	0.002	Equivalent
Strict classical random features	+2.27	$[+1.92, +2.63]$	0.999	1.000	PCAE2D better
Random VQC	+6.97	$[+5.56, +8.39]$	1.000	1.000	PCAE2D better

$[-0.91, -0.46]$ dB and Wilcoxon $p = 0.0625$ at $n = 5$ seeds. The same numerical direction holds on five further real datasets. The tested PCAE map is therefore informative, and on ordinary held-out reconstruction it stays within 1 dB of the simpler maps.

All controls are matched at the readout. Each passes a 32-D feature vector to the same linear readout with 2112 parameters, on the same splits. This matches the readout resources. It does not match the feature class or the compute cost. The controls differ in what their features can express. The no-entanglement and ZZ-only controls remove all or part of the entangling pair channel, and the quadratic classical control adds second-order classical terms. The compute needed to produce the features is not matched, since PCAE2D evaluates a quantum circuit for every patch. The held-out PCAE2D map is the 32-D variant of Table S-resource_capacity, with six pair channels and 14 CNOTs, rather than the 19-D map of Table 1. The quadratic classical control augments the 18 local features with the nine products of adjacent local-feature pairs and their sines, together with the positional features, selected to the same 32-D width.

A TOST equivalence test (Schuirmann 1987; Lakens 2017) at a declared ± 1 dB margin confirms that this gap is small enough to call the two *competitive* rather than merely “not shown to differ”. The estimand is the seed-level PCAE2D-minus-control PSNR difference, averaged over held-out images within each seed. Equivalence requires both one-sided tests to reject at $\alpha = 0.05$. This is the same as the 90% confidence interval lying inside $[-1, +1]$ dB (Table 4). The margin is a declared choice. With five seeds, $p = 0.0625$ is the smallest attainable nonzero two-sided Wilcoxon signed-rank p -value, so that test cannot reject at the 0.05 level. The t -based 90% interval is wider than the percentile-bootstrap 95% interval in these data; bootstrap intervals can be unstable with only five seed-level observations. For scale, a 1 dB PSNR difference corresponds to a factor of approximately $10^{0.1} \simeq 1.26$ in MSE. After Holm correction across the seven comparisons, four controls meet the equivalence criterion against PCAE2D within that margin: no entanglement, ZZ-only observables, quadratic classical features, and shuffled pair observables. The raw/local linear control meets the uncorrected criterion ($p = 0.034$) but not the Holm-corrected criterion (adjusted $p = 0.103$). The two weakest controls, strict classical random features and random-VQC, are not equivalent because PCAE2D substantially outperforms them.

We report this as a complement to the positive pair-observable and symmetry results below. It delineates their tested boundary and records the provenance checks that prevented exploratory artifacts from becoming claims.

A matched-budget, multi-seed rerun of the component-ablation table reaches the same conclusion on ordinary per-image reconstruction. The rerun uses 240 steps and 3 seeds per variant. A plain classical-replacement control ties or slightly exceeds the quantum baseline, and only a circuit-free random-VQC control performs clearly worse. This rerun uses a descriptive pooled-standard-deviation comparison rule at 3 seeds rather than a calibrated test, and full results are in the companion supplement. Section 5.2 discusses why this ties-with-classical result and the pair-observable necessity result of Section 4.8 are consistent.

4.4 A real hardware reconstruction pilot

The results above are noiseless-simulator results. We now report a small but complete image-reconstruction pilot executed on real IBM Quantum hardware (`ibm_fez`, a 156-qubit Heron-class processor), going beyond the compiled-circuit feasibility probe documented in the supplement.

The pilot covers five images. These are the full 16-patch *Monalisa* benchmark and four native-resolution CIFAR-10 images (*cat*, *ship*×2, *airplane*), each optimised directly on its own target image following the same per-image protocol as Sections 4.1 and 4.2. For each image we took an already-trained PCAE1D/PCAE2D checkpoint’s exact learned parameters and rebuilt its measurement circuits gate-for-gate in Qiskit. We verified the translation against the original PennyLane simulator to within numerical noise, at most 7.9×10^{-4} maximum absolute error per observable across every patch, with full validation numbers in the companion supplementary document. We then executed the same circuits on real hardware with 256 shots per measurement basis. This is a fixed per-basis shot budget, rather than an adaptive shadow-tomography-style estimation scheme (Huang et al. 2020). The hardware-measured observables were decoded by the same trained classical decoder, with no retraining and no simulator involved anywhere in the reconstruction path.

Reconstructions from real, noisy hardware measurements retain recognisable image structure. *Monalisa* reaches 25.90 dB PSNR from hardware measurements, against a 37.99 dB noiseless-simulator replay of the identical circuit (Figure 2). The four CIFAR-10 images average 29.56 ± 2.03 dB on hardware against 42.69 ± 3.84 dB in simulation (Figure 3, Table 5). The observed gap is consistent with noise in these unmitigated device executions, although the five-image pilot does not isolate individual noise sources. Circuits were transpiled at optimisation level 1 and submitted through SamplerV2 with default options; dynamical decoupling, Pauli twirling and readout-error mitigation were not enabled. Each image used 256 shots per basis, totalling 12,288 shots for *Monalisa* and 8,192 for each CIFAR-10 image. The original jobs’ physical-qubit layouts and contemporaneous calibration timestamps were not serialised and could not subsequently be recovered. Every hardware reconstruction remains above the random-latent control range (11–15 dB in the companion supplementary study). The pilot used 176 circuits; the contemporaneous account-meter reading increased

Table 5 Real IBM hardware reconstruction pilot (backend `ibm_fez`, 256 shots/basis, no retraining). PSNR compares hardware-decoded and simulator-decoded reconstructions against the same ground-truth image.

Image	Patches	Sim. PSNR (dB)	HW PSNR (dB)
Monalisa (PCAE1D)	16	37.99	25.90
CIFAR cat (PCAE2D)	16	44.85	31.52
CIFAR ship, hydrofoil (PCAE2D)	16	36.56	26.44
CIFAR ship, sea boat (PCAE2D)	16	42.57	31.20
CIFAR airplane (PCAE2D)	16	46.79	29.10
CIFAR mean $\pm$ std (4 images)	—	42.69 ± 3.84	29.56 ± 2.03



Fig. 2 *Monalisa* reconstructed from real IBM Quantum hardware measurements (right) against the same trained checkpoint’s noiseless-simulator replay (centre) and ground truth (left). The hardware image is visibly noisier and exhibits patch-boundary banding, while the subject remains recognisable.

by 16 quantum-processor seconds, with the provenance limitation documented in the supplement.

The pilot establishes that the trained PCAE observable channel survives execution on present-day hardware. With five images it is a pilot, and it makes no hardware-advantage claim. The learned parameters, the measurement circuits and the classical decoder all transfer unchanged, and the decoded images remain quantitatively informative at 256 shots per basis on an unmitigated device. This is the sense in which PCAE runs on real quantum hardware. The companion supplementary document gives the full methodology, covering exact gate sequences, per-basis measurement construction, job IDs, and pre-submission local-validation results.

The same checkpoint-replay methodology supports a further hardware replay, reported in the companion Supplementary Material. Fresh PCAE1D checkpoints were trained at $N \in \{4, 6, 8\}$ and replayed epoch-by-epoch on real IBM Quantum hardware and on the IonQ cloud simulator (IonQ 2026). That exercise serves the training-dynamics diagnostic of Section 4.9. It is reported in the Supplementary Material as an interpretability readout: it is a single-patch, single-seed probe, and the threshold it tracks appears with and without the Hamiltonian term (Section 4.9).

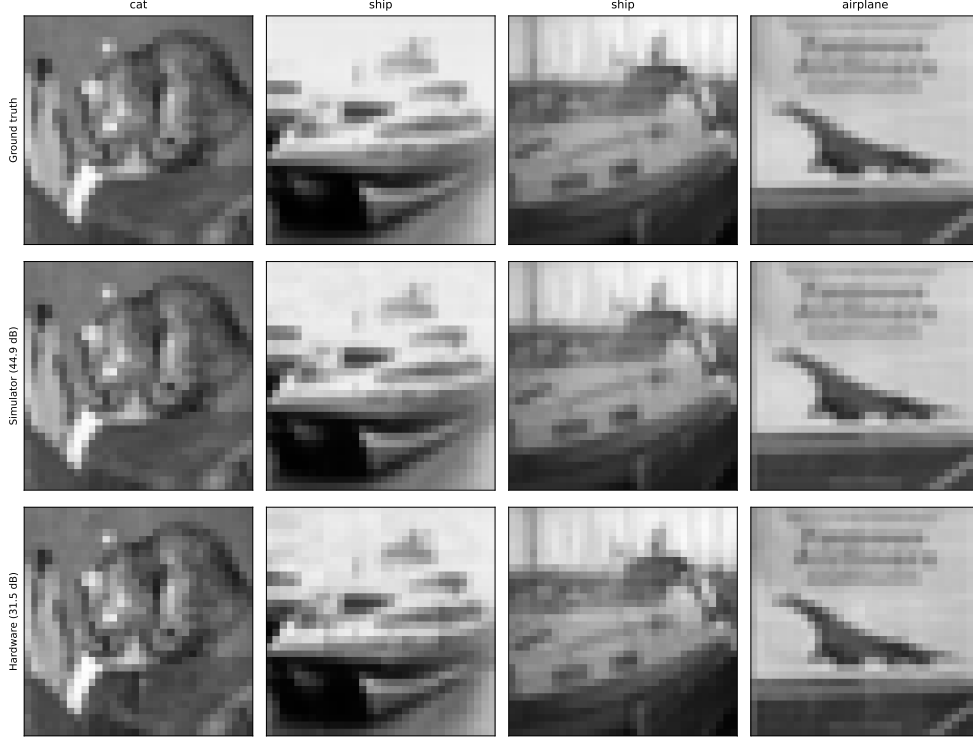


Fig. 3 Four CIFAR-10 images reconstructed from real IBM Quantum hardware measurements (bottom row) against noiseless-simulator replays of the same trained PCAE2D checkpoints (middle row) and ground truth (top row).

4.5 Noise/shot trade-off: a free simulator complement

The five-image pilot above used a single shot budget (256/basis) and one execution per image, which bounds how much it can say about noise sensitivity on its own. To use IBM Quantum time well, we separate what a *calibrated device-noise simulation* can establish at zero hardware cost from what genuinely requires hardware. Using `qiskit-aer` with the live calibration snapshot of `ibm_fez` (**FakeFez**, no quantum-processor time consumed), we replay the same already-trained checkpoints’ circuits at four shot budgets (256/512/1024/4096 shots per basis, three repeated executions each) and compare against both the noiseless-simulator ceiling and the real-hardware point already reported above.

Two findings follow, averaged over the five checkpoints. First, shot noise accounts for only part of the noiseless-to-real gap. Moving from the noiseless ceiling to a calibrated-noise simulation at the same 256-shot budget already costs 8.68 dB on average, before any real device is involved. Second, more shots do not reliably recover it. Increasing to 4096 shots per basis changes PSNR by only -0.05 dB on average across the five checkpoints, with individual images ranging from a 1.33 dB gain to a 0.96 dB

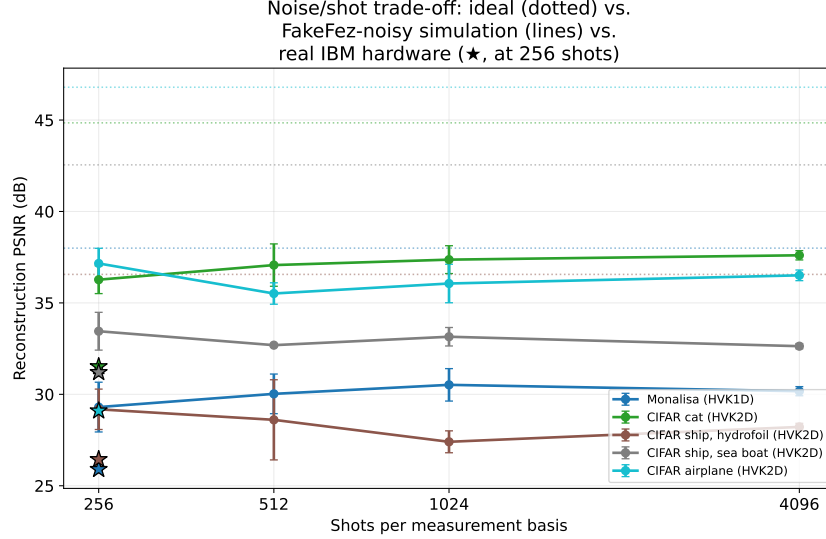


Fig. 4 Calibrated device-noise simulation (FakeFez, mean $\pm$ std over three executions). Dotted lines show noiseless ceilings and stars show real-hardware measurements. Increasing 256 to 4096 shots changes mean PSNR by only -0.05 dB.

loss. At this circuit size and shot range, shot count is not a lever that reliably improves reconstruction quality.

Comparing the 256-shot noisy simulation directly against the real-hardware point isolates what the calibration snapshot does *not* capture. That comparison leaves a further 4.24 dB average gap, ranging from 2.25 to 8.06 dB across images. This residual is consistent with effects that the calibration model does not capture, such as drift between calibration and execution, crosstalk, and correlated errors. The pilot does not separate these sources. The airplane image is the clear outlier, with an 8.06 dB residual gap, roughly double any other image. We report it rather than exclude it, since a single-execution, five-image pilot cannot distinguish an image-specific effect from ordinary job-to-job variation. This calibrated-noise simulation is a free, repeatable complement to the pilot, not a substitute for it. It bounds how much of the observed hardware gap is attributable to modelled device noise rather than unmodelled effects, using zero additional quantum-processor time.

4.6 Real-hardware anchor points: repeated execution and a second backend

The simulated trade-off above is useful only if it reflects real devices. We checked this directly on hardware with four jobs from the free-tier IBM Quantum allowance. These four jobs used well under 1% of the monthly limit. Two checkpoints, *Monalisa* and *CIFAR cat*, were replayed with no retraining on `ibm_marrakesh` and `ibm_kingston`. Both backends are distinct from the original pilot’s `ibm_fez`. We ran them at the original 256-shot budget and, separately, at 1024 shots.

Table 6 Real-hardware anchor points: same checkpoints, no retraining, on two further backends. The original pilot’s `ibm_fez` point is shown for reference.

Image	Backend	Shots	PSNR (dB)
Monalisa	<code>ibm_fez</code> (original)	256	25.90
Monalisa	<code>ibm_marrakesh</code>	256	25.94
Monalisa	<code>ibm_kingston</code>	1024	26.10
CIFAR cat	<code>ibm_fez</code> (original)	256	31.52
CIFAR cat	<code>ibm_kingston</code>	256	28.93
CIFAR cat	<code>ibm_kingston</code>	1024	31.24

Monalisa reproduces almost exactly across backends at matched shots (25.94 vs. 25.90 dB, `ibm_marrakesh` vs. `ibm_fez`), and gains only 0.16 dB from quadrupling shots to 1024, consistent with the simulated finding above that shot count is not a reliable lever. The CIFAR *cat* point is the more informative one. On `ibm_kingston` at the original 256-shot budget it reaches 28.93 dB, which is 2.59 dB below the `ibm_fez` pilot point. Its own 1024-shot execution on the same backend recovers to 31.24 dB, within 0.28 dB of the original. This is real job-to-job and backend-to-backend variation of a magnitude comparable to the unmodelled residual identified by the simulator comparison (Section 4.5). It gives direct evidence that repeated execution matters at least as much as backend choice or shot count. A single 256-shot run on one backend is not yet a stable estimate of hardware performance.

4.7 Exact D_4 -equivariant observable pooling

A D_4 -pooled PCAE2D observable-grid map is equivariant to the eight symmetries of the square *by construction*:

$$\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx), \quad \Phi_{D_4}(hx) = P_h \Phi_{D_4}(x) \quad \forall h \in D_4, \quad (4)$$

Here D_4 acts on an image x by rotating or reflecting its 4×4 grid of patches, and P_g permutes the per-patch observable vectors in the same way, so that $P_{gh} = P_g P_h$. The property then holds exactly for any base map Φ . Substituting hx and re-indexing the sum with $g' = gh$ gives $P_g^{-1} = P_h P_{g'}^{-1}$, and hence $\Phi_{D_4}(hx) = P_h \Phi_{D_4}(x)$. The numerical check below therefore tests the implementation, not the property. We run it over 7000 patch transforms of cached CIFAR images (Table 7, Figure 5). The pooled map’s equivariance error is $9.57 \times 10^{-17} \pm 1.26 \times 10^{-17}$, which is floating-point noise. The unpooled positional map has an error of 0.815 ± 0.129 , and the local/raw control has 0.837 ± 0.131 . Both are of order one. The patch grid’s geometry therefore makes the unpooled map *D_4 -motivated* only, and the explicit pooling construction is what delivers the property. This construction operates at the feature-grid level and serves here as a numerical diagnostic, rather than as a component of the trainable reconstruction pipeline. Integrating it end-to-end (Section 5.4) is concrete future work. This route to

Table 7 D_4 equivariance error for unpooled and pooled observable-grid maps (lower is better; the pooled map is group-averaged over all eight square symmetries).

Model	Mean error	Std. dev.
D_4 -pooled PCAE2D	9.57×10^{-17}	1.26×10^{-17}
No positional encoding	7.40×10^{-1}	1.52×10^{-1}
PCAE2D positional	8.15×10^{-1}	1.29×10^{-1}
Local/raw	8.37×10^{-1}	1.31×10^{-1}

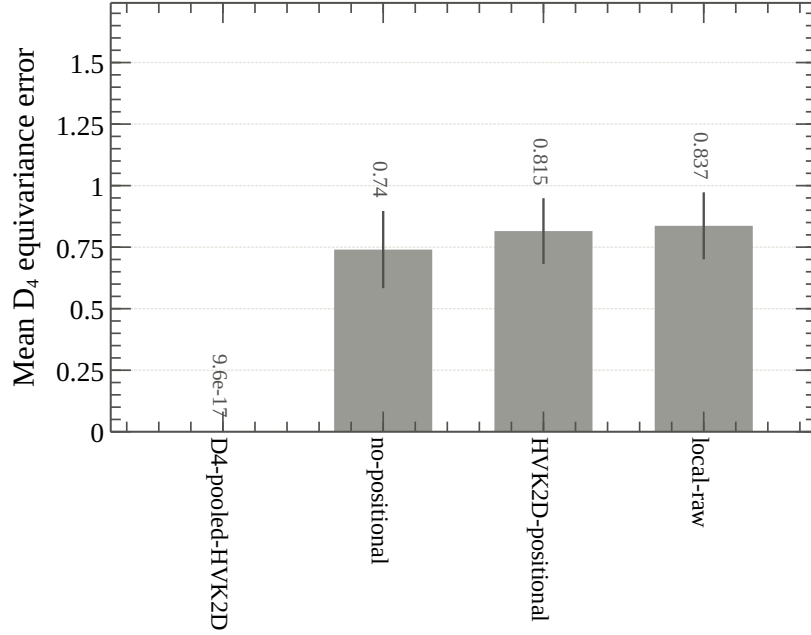


Fig. 5 D_4 equivariance diagnostic. The group-pooled map is equivariant to numerical precision. The unpooled positional map is not.

symmetry differs from equivariant quantum neural networks, in which every circuit layer is built to commute with the group action (Nguyen et al. 2024). A recent image classifier of that kind builds C_4 rotation symmetry into the circuit itself (San Sebastian Sein et al. 2025). In PCAE the symmetry comes from averaging the observable grid over the group after measurement. The circuit itself is not equivariant, and the same average applies equally to a classical feature map.

Table 8 PCAE2D restricted pair-correlation diagnostic across five seeds. Deliberately favourable to pair-observable structure; evidence for pair-observable necessity under a matched linear readout.

Model	Mean MSE	Mean PSNR (dB)	Mean R^2
PCAE2D pair observables	8.5969×10^{-4}	32.77	0.9735
Freeze quantum	4.7904×10^{-3}	27.34	0.8533
No pair observables	3.1461×10^{-2}	15.03	0.0191
Raw-linear classical	3.1654×10^{-2}	15.00	0.0133
Parameter-matched classical	3.3033×10^{-2}	14.82	-0.0297
Random VQC	3.7616×10^{-2}	14.25	-0.1732
Freeze classical	6.0973×10^{-2}	12.17	-0.9025

4.8 Pair-observable necessity in the nonlocal-structure protocol

The diagnostic tests whether a pair-observable channel is needed to fit a target that depends on pair correlations. The target is a fixed synthetic function of *simulated* patch features, and those features are not concatenated into any candidate’s own input block. The companion supplement gives the leakage-audit procedure that rules out circularity. Every control receives the same raw inputs as PCAE2D. The experiment demonstrates representational capability under a deliberately aligned protocol, not a general performance advantage.

Table 8 and Figure 6 report the result across five seeds. PCAE2D’s pair-observable channel reaches mean $R^2 = 0.9735$, or 32.77 dB PSNR. This is a 17.74 dB improvement over a no-pair-observables control ($R^2 = 0.019$) and 18.52 dB over random-VQC latents ($R^2 = -0.173$). Freezing the quantum circuit, so that features are fixed and only the readout is trained, still reaches $R^2 = 0.853$. A raw-linear classical control ($R^2 = 0.0133$), a parameter-matched classical control ($R^2 = -0.030$), a random-VQC control ($R^2 = -0.173$), and a frozen-classical control ($R^2 = -0.903$) all fail on this deliberately pair-dependent target. The no-pair-observables control retains only single-site nonlinear channels and removes every cross-site product feature. Under the fixed linear readout, it therefore cannot construct the cross terms on which the target depends.

The mechanism is explicit under the fixed linear readout. A linear map of single-patch expectations $\{\langle O_i \rangle\}$ cannot form the cross-term $\langle O_i \rangle \langle O_j \rangle$ that the target depends on. The pair measurement supplies the joint correlator $\langle O_i O_j \rangle$ directly as a feature. Formally, let $z_i = \langle O_i \rangle$ and let every admissible predictor be affine, $f(z) = w^\top z + b$. Then $\partial^2 f / \partial z_i \partial z_j = 0$ for all i, j . A target with a term $c z_i z_j$, $c \neq 0$, has this mixed derivative equal to c wherever z_i and z_j vary independently, so no affine f can fit it. Adding $z_i z_j$ to the feature vector makes the term linearly representable. A pair-observable channel is therefore *representationally necessary within the tested feature class and fixed linear-readout protocol* to fit this target. A classical model that forms the same products, for example through quadratic features or a nonlinear readout, would also fit this target. For this reason we scope the claim to the linear protocol rather than to classical models in general.

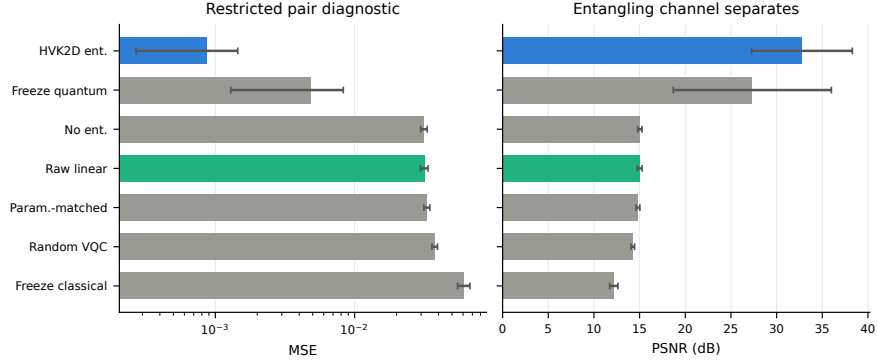


Fig. 6 Restricted pair-correlation diagnostic summary (mean $\pm$ std over five seeds). The pair-observable feature map dominates every control on this deliberately favourable task.

4.9 A scoped training-dynamics diagnostic

Beyond the results above, we tracked two exploratory training-dynamics readouts. These are a magnetisation-style statistic $M_z(t)$ and a signed energy-to-entanglement ratio $R_{ES}(t)$. Both drift smoothly over training with run-to-run fluctuations, and neither shows a sharp transition. Any apparent onset is a within-run median-plus- 2σ threshold crossing, rather than a calibrated statistical test.

A negative control fixes the scope. We applied the identical protocol to a classical-replacement variant, in which the Hamiltonian energy is identically zero throughout training. That variant crosses the threshold in all 4/4 tested cases, which is as often as the Hamiltonian-regularised model’s 4/4. The threshold therefore fires whether the Hamiltonian term is present or absent, and it does not track that term. We therefore report these traces as an interpretability readout on the learned energy and entanglement. The companion Supplementary Material gives the full protocol and tables.

5 Discussion

5.1 Effect of the Hamiltonian energy term

In this single-seed comparison, the signed Hamiltonian-energy objective substantially reduces reconstruction quality: removing it increases PSNR from 38.63 to 44.56 dB (Table 9). The legacy objective adds a signed mean energy to the reconstruction loss. Because learned energy can go negative, the optimiser can lower total loss by driving energy down without improving reconstruction. A contrastive variant, assigning lower energy to correct feature-position pairs than to shuffled pairs, recovers part of the gap (42.92 dB) but stays below its no-energy counterpart. These are fresh same-environment measurements (*Monalisa*, 200 steps, seed 42), and they supersede earlier values produced by a script that cannot be audited.

Table 9 Hamiltonian-sector controls (single-seed, fresh same-environment reproduction: *Monalisa*, 200 steps, seed 42). The energy objective is diagnostically useful but does not improve reconstruction accuracy. See the Supplementary Material for the full reproducibility investigation and additional observable-sector controls.

Experiment	PSNR (dB)	Interpretation
Legacy baseline, signed energy	38.63	Standard energy-regularised run
No energy loss	44.56	Removing energy improves PSNR
Contrastive Hamiltonian core	42.92	Stronger objective helps modestly vs. baseline, still below no-energy-loss

The learned couplings do not enter the decoder directly, but the energy term contributes gradients during training and therefore changes the learned circuit representation. It is consequently an auxiliary training objective as well as a post-training diagnostic. A physically motivated regulariser has room to help when it resolves an otherwise underdetermined objective. Under this protocol, the reconstruction loss alone provides the stronger optimisation signal. The Supplementary Material gives the full account.

The training-dynamics readouts of Section 4.9 provide a descriptive window into how the learned energy and entanglement evolve. They are descriptive readouts rather than calibrated statistical tests, and the thermodynamic reading is outside their scope.

5.2 Task-dependent representational scope

Whether a quantum feature map offers any advantage over a classical model is a property of the specific data and target, not a fixed property of the model class. Dequantisation results have repeatedly shown that apparent quantum advantages on structured, low-entanglement classical data can be matched once the right classical feature is identified (Huang et al. 2021; Tang 2019). Where provable separations do exist, they are established for carefully constructed learning problems rather than for generic structured data (Gyurik and Dunjko 2023). We separate three questions. First, PCAE at six qubits can be simulated exactly on a classical computer, which says little about larger scales. Second, whether its model class admits an efficient classical replacement at scale is open. Tensor-network analyses give conditions for such dequantisation of variational models (Shin et al. 2024), and a classical tie at small scale does not settle it. Random Fourier features can also replace some variational models efficiently, under stated conditions (Sweke et al. 2025). Quantum convolutional networks on common image benchmarks can be simulated efficiently by a classical method based on Pauli shadows (Bermejo et al. 2026). Third, on the tested ordinary reconstruction tasks, PCAE stays within the declared 1 dB margin of the stronger matched classical controls. This matches large benchmarks in which simple classical models often match or beat small quantum models, and removing entanglement often costs little (Bowles et al. 2024). In PCAE the quantum processor evaluates only the circuit and returns the vector of measured Pauli expectation values. The MPS compression is quantum-inspired but runs on classical hardware, as do the Fourier positional encoding, the D_4

pooling, the decoder and the optimisation. Any quantum contribution to PCAE must therefore enter through the measured observable vector.

The restricted diagnostic of Section 4.8 is constructed so that the relevant structure is a genuine distant-element product, which raw or local features cannot represent under a fixed linear readout. That is the regime in which PCAE’s pair-observable channel separates. Natural image patches at the resolution and patch size used here are low-entanglement, locally structured data, where a linear or shallow classical readout is already close to sufficient. The distinction lies in the measured pair-observable channel rather than in the circuit. The ansatz uses strongly-entangling layers and does generate an entangled state, but whether the measured channel carries task-relevant information is a property of the target. On natural-image targets the pair observables add little beyond the local ones. On the constructed nonlocal target they are necessary. Together with the held-out comparisons of Section 4.3, these results delineate where the tested entangling representation does and does not yield a measurable benefit.

Three of PCAE’s ingredients recur individually in the hybrid quantum-vision literature, covering MPS compression of structured input (Stoudenmire and Schwab 2016; Han et al. 2018; Cirac et al. 2021; Guala et al. 2023), a Pauli-correlator latent space handed to a classical decoder (Havlíček et al. 2019), and a physically motivated auxiliary regulariser alongside a task loss (Shi et al. 2023). Its specific instantiation, a six-qubit VQC with a 19–27-D correlator vector and a graph-Hamiltonian term, sits within the parameter ranges typical of that literature. The measured behaviour is therefore evidence about the design pattern itself. We do not claim the boundary transfers unchanged to every dataset, patch size, or qubit count.

5.3 Topology alignment as an inductive bias

Aligning the latent interaction graph with image-patch adjacency (PCAE2D vs. PCAE1D) is an architectural inductive bias, and the companion supplementary study tests it directly under matched held-out conditions. Its real-circuit subset (3 seeds, 3 held-out CIFAR-10 images) gives a PCAE1D-minus-PCAE2D mean PSNR difference of 0.16 dB. After averaging images within seed, the bootstrap 95% CI is $[-0.38, 0.46]$ dB and the seed-level Wilcoxon test gives $p = 0.50$. A broader five-seed surrogate comparison across CIFAR-10, Fashion-MNIST, and PathMNIST likewise finds numerically small topology differences, at most 0.07 dB in mean PSNR. PCAE2D completes the matched 90-step runs in roughly half the wall-clock time. At the tested scale, topology is therefore primarily a resource-cost choice rather than a reliable accuracy advantage. These tests do not establish formal equivalence.

Table 10 contrasts PCAE’s structural properties against reference frameworks, including the three nearest hybrid-vision architecture families. Its distinguishing features are a Pauli-correlator latent space, a 2D Fourier positional bias with enforced D_4 pooling, and a Hamiltonian energy diagnostic, each documented and scoped in the preceding sections.

Table 10 Architectural differentiation from representative baseline designs, including the three nearest hybrid-vision architecture families (register-based QAE, quantum-kernel vision, TN-circuit classifier).

Axis	Conv. AE baseline	GAN baseline	Register-based QAE	Quantum-kernel vision	TN-circuit classifier	PCAE (ours)
Latent representation	Continuous $\mathbb{R}^d$	Continuous $\mathbb{R}^d$	Quantum register	Scalar kernel value	MPS/circuit state	Pauli-correlator vector
Spatial bias	Convolutional	Convolutional	Encoding-dependent	Encoding-dependent	MPS chain ordering	2D Fourier encoding
Auxiliary objective	Bottleneck / reconstruction	Adversarial loss	Fidelity / reconstruction	None (kernel value only)	Classification loss	Hamiltonian diagnostic
Enforced group symmetry	Not in baseline	Not in baseline	Not in reference design	Not in reference design	Not in reference design	D_4 observable pooling
Evaluated role	Reconstruction	Reconstruction	Compression	Classification	Classification	Reconstruction + diagnostics

5.4 Validated scope

- Every reconstruction result in this paper is per-image trained, except the held-out comparison of Section 4.3. Only that held-out comparison, reported in full in the companion supplement, supports a generalisation claim.
- The D_4 -pooled map (Section 4.7) is a feature-grid-level construction and numerical diagnostic. Integrating it into the trainable end-to-end pipeline is future work (Section 6).
- The pair-observable necessity result (Section 4.8) is scoped to a task constructed to require nonlocal structure. Ordinary natural-image reconstruction lies outside that scope (see Section 5.2).
- The hardware pilot (Section 4.4) is a five-image, unmitigated pilot with one run per image on the main backend rather than a statistically resolved hardware benchmark.
- The training-dynamics traces (Section 4.9) are an interpretability readout. The threshold fires whether the Hamiltonian term is present or absent, at 4/4 in both cases, with the full table in the Supplementary Material.
- The matched real-circuit topology comparison uses 3 seeds and 3 held-out images per topology, and the broader five-seed comparison is surrogate-based. These results support a resource-cost interpretation rather than formal equivalence of the topologies.
- Every result here is at a six-qubit, depth-18 scale. Trainability of variational models is known to degrade with width and depth through barren-plateau effects (McClean

et al. 2018; Larocca et al. 2025), so whether the tested boundary moves at larger scale is an open question.

6 Conclusion

We presented the Pauli-Correlator Autoencoder (PCAE), a hybrid quantum-classical architecture for image reconstruction that assembles matrix-product-state patch features, a measured Pauli-correlator latent space, enforced D_4 -equivariant observable pooling, and a learnable Hamiltonian energy diagnostic into a single pipeline (Section 5.2). Across six real image datasets, per-image-trained PCAE2D models reach 25.8–41.6 dB PSNR. On a resource-matched, multi-seed held-out comparison, a TOST equivalence test shows PCAE2D is *competitive with*, rather than inferior or superior to, simple classical controls (Section 4.3). A hardware pilot replayed trained checkpoints on IBM Quantum devices without retraining, establishing feasibility at a five-image scope with one run per image on the main backend (Section 4.4).

Pair observables are necessary on a task requiring nonlocal structure, under a fixed linear readout (Section 4.8). The Hamiltonian energy term serves as an interpretable diagnostic of the trained representation (Section 5.1). The pipeline executes natively on real quantum hardware. It is also exactly D_4 -equivariant by construction, which we report as a design-correctness property rather than a quantum capability, since the same Reynolds averaging applied to a classical feature map is equally exact. We make no claim of quantum advantage. At the six-qubit scale tested here, PCAE is competitive with resource-matched classical baselines and runs on real quantum hardware.

Future work should integrate the D_4 -pooled map into an end-to-end trainable pipeline (Section 5.4), scale the hardware pilot toward a statistically resolved benchmark with error mitigation and repeated trials, search for further real-world tasks with nonlocal structure where PCAE’s pair-observable separation translates into a reconstruction advantage, and extend the resource-matched, leakage-audited evaluation protocol to other hybrid quantum vision architectures.

Supplementary information. A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, accompanies this manuscript.

Acknowledgements. We thank the maintainers of the open-source scientific computing libraries used in this work.

Statements and Declarations

Competing interests. The authors declare no competing interests.

Funding. No funding was received for conducting this study.

Data availability. Dataset provenance and access routes are cited in the manuscript and detailed in the supplementary artifact map. All datasets used (CIFAR-10, MNIST, Fashion-MNIST, and the PathMNIST, BloodMNIST and PneumoniaMNIST subsets of MedMNIST v2) are publicly available.

Code availability. The source code, experiment drivers, machine-readable result summaries, validation arrays, and figure inputs supporting this study are included in the accompanying repository. The file `project_artifacts/submission_claim_audit.json` maps each numerical headline claim to its retained source artifact, covering the resource-matched held-out comparison, on which the tested PCAE map is competitive with rather than superior to the classical controls, and the withdrawn training-dynamics material. **[Student: Section 4.9 still reports the training-dynamics traces as an interpretability readout. Please say which training-dynamics material was withdrawn, so that the two statements agree.]** Automated tests verify those mappings. IBM Quantum job identifiers and reconstructed outputs are retained; as disclosed in the supplement, the raw account-usage service response was not serialised, so the reported quota change is an execution-log record rather than a repository-recomputable measurement.

Author contributions. **Sparsho Chakraborty:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Siddhartha Patra:** Conceptualization, Supervision, Writing – review & editing, Resources. Both authors reviewed and approved the final manuscript.

References

- Araz JY, Spannowsky M (2022) Classical versus quantum: comparing tensor-network-based quantum circuits on Large Hadron Collider data. *Physical Review A* 106(6):062423. <https://doi.org/10.1103/PhysRevA.106.062423>
- Asaoka H, Kudo K (2025) Quantum autoencoders for image classification. *Quantum Machine Intelligence* 7(2):71. <https://doi.org/10.1007/s42484-025-00297-x>
- Bauer B, Bravyi S, Motta M, et al (2020) Quantum algorithms for quantum chemistry and quantum materials science. *Chemical Reviews* 120(22):12685–12717. <https://doi.org/10.1021/acs.chemrev.9b00829>
- Beaulieu D, Miracle D, Pham A, et al (2022) Quantum kernel for image classification of real world manufacturing defects. *arXiv preprint arXiv:2212.08693* <https://doi.org/10.48550/arXiv.2212.08693>
- Bermejo P, Braccia P, Rudolph MS, et al (2026) Quantum convolutional neural networks are effectively classically simulable. *PRX Quantum* 7:020304
- Bowles J, Ahmed S, Schuld M (2024) Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv preprint arXiv:2403.07059*
- Bronstein MM, Bruna J, Cohen T, et al (2021) Geometric deep learning: grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478* <https://doi.org/10.48550/arXiv.2104.13478>

- Cerezo M, Arrasmith A, Babbush R, et al (2021) Variational quantum algorithms. *Nature Reviews Physics* 3(9):625–644. <https://doi.org/10.1038/s42254-021-00348-9>
- Chakraborty S, Mandal SB, Shaikh SH (2022) Quantum image processing: challenges and future research issues. *International Journal of Information Technology* 14(1):475–489. <https://doi.org/10.1007/s41870-018-0227-8>
- Cirac JJ, Pérez-García D, Schuch N, et al (2021) Matrix product states and projected entangled pair states: concepts, symmetries, theorems. *Reviews of Modern Physics* 93(4):045003. <https://doi.org/10.1103/RevModPhys.93.045003>
- Cohen T, Welling M (2016) Group equivariant convolutional networks. In: *Proceedings of the 33rd International Conference on Machine Learning, Proceedings of Machine Learning Research*, vol 48. PMLR, pp 2990–2999
- Cong I, Choi S, Lukin MD (2019) Quantum convolutional neural networks. *Nature Physics* 15(12):1273–1278. <https://doi.org/10.1038/s41567-019-0648-8>
- Dupont E, Goliński A, Alizadeh M, et al (2021) COIN: COmpression with implicit neural representations. *arXiv preprint arXiv:2103.03123*
- Fujihashi T, Koike-Akino T (2024) Quantum implicit neural compression. *arXiv preprint arXiv:2412.19828*
- Guala D, Zhang S, Cruz E, et al (2023) Practical overview of image classification with tensor-network quantum circuits. *Scientific Reports* 13(1):4427. <https://doi.org/10.1038/s41598-023-30258-y>
- Gyurik C, Dunjko V (2023) Exponential separations between classical and quantum learners. *arXiv preprint arXiv:2306.16028* <https://doi.org/10.48550/arXiv.2306.16028>
- Haga A (2024) Quantum annealing-based computed tomography using variational approach for a real-number image reconstruction. *Physics in Medicine and Biology* 69(4):04NT02. <https://doi.org/10.1088/1361-6560/ad2155>
- Han ZY, Wang J, Fan H, et al (2018) Unsupervised generative modeling using matrix product states. *Physical Review X* 8(3):031012. <https://doi.org/10.1103/PhysRevX.8.031012>
- Havlíček V, Córcoles AD, Temme K, et al (2019) Supervised learning with quantum-enhanced feature spaces. *Nature* 567(7747):209–212. <https://doi.org/10.1038/s41586-019-0980-2>
- Huang HY, Kueng R, Preskill J (2020) Predicting many properties of a quantum system from very few measurements. *Nature Physics* 16(10):1050–1057. <https://doi.org/10.1038/s41567-020-0932-7>

- Huang HY, Broughton M, Mohseni M, et al (2021) Power of data in quantum machine learning. *Nature Communications* 12(1):2631. <https://doi.org/10.1038/s41467-021-22539-9>
- Hur T, Kim L, Park DK (2022) Quantum convolutional neural network for classical data classification. *Quantum Machine Intelligence* 4(1):3. <https://doi.org/10.1007/s42484-021-00061-x>
- IonQ (2026) Backends. IonQ Quantum Cloud Documentation, URL <https://docs.ionq.com/user-manual/backends>, accessed 24 July 2026
- Krizhevsky A (2009) Learning multiple layers of features from tiny images. Tech. rep., University of Toronto, Toronto, Ontario
- Lakens D (2017) Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science* 8(4):355–362. <https://doi.org/10.1177/1948550617697177>
- Larocca M, Thanasilp S, Wang S, et al (2025) Barren plateaus in variational quantum computing. *Nature Reviews Physics* 7:174–189. <https://doi.org/10.1038/s42254-025-00813-9>
- Latorre JI (2005) Image compression and entanglement. arXiv preprint arXiv:quant-ph/0510031 <https://doi.org/10.48550/arXiv.quant-ph/0510031>
- Le PQ, Dong F, Hirota K (2011) A flexible representation of quantum images for polynomial preparation, image compression, and processing operations. *Quantum Information Processing* 10:63–84. <https://doi.org/10.1007/s11128-010-0177-y>
- LeCun Y, Bottou L, Bengio Y, et al (1998) Gradient-based learning applied to document recognition. *Proceedings of the IEEE* 86(11):2278–2324. <https://doi.org/10.1109/5.726791>
- Liu Z, Shen PX, Li W, et al (2023) Quantum capsule networks. *Quantum Science and Technology* 8(1):015016. <https://doi.org/10.1088/2058-9565/aca55d>
- Lizzio Bosco D, Wang S, Serra G, et al (2026) Quantum implicit neural representations for novel view synthesis. arXiv preprint arXiv:2601.05250
- McClellan JR, Boixo S, Smelyanskiy VN, et al (2018) Barren plateaus in quantum neural network training landscapes. *Nature Communications* 9(1):4812. <https://doi.org/10.1038/s41467-018-07090-4>
- Meyer JJ, Mularski M, Gil-Fuster E, et al (2023) Exploiting symmetry in variational quantum machine learning. *PRX Quantum* 4(1):010328. <https://doi.org/10.1103/PRXQuantum.4.010328>

- Mildenhall B, Srinivasan PP, Tancik M, et al (2020) NeRF: Representing scenes as neural radiance fields for view synthesis. In: Computer Vision – ECCV 2020, Lecture Notes in Computer Science, vol 12346. Springer, pp 405–421, https://doi.org/10.1007/978-3-030-58452-8_24
- Nau MA, Vija AH, Gohn W, et al (2023) Exploring the limitations of hybrid adiabatic quantum computing for emission tomography reconstruction. *Journal of Imaging* 9(10):221. <https://doi.org/10.3390/jimaging9100221>
- Nguyen QT, Schatzki L, Braccia P, et al (2024) Theory for equivariant quantum neural networks. *PRX Quantum* 5(2):020328. <https://doi.org/10.1103/PRXQuantum.5.020328>
- Preskill J (2018) Quantum computing in the NISQ era and beyond. *Quantum* 2:79. <https://doi.org/10.22331/q-2018-08-06-79>
- Ran SJ, Sun ZZ, Fei SM, et al (2020) Tensor network compressed sensing with unsupervised machine learning. *Physical Review Research* 2(3):033293. <https://doi.org/10.1103/PhysRevResearch.2.033293>
- Romero J, Olson JP, Aspuru-Guzik A (2017) Quantum autoencoders for efficient compression of quantum data. *Quantum Science and Technology* 2(4):045001. <https://doi.org/10.1088/2058-9565/aa8072>
- San Sebastian Sein P, Cañizo M, Orús R (2025) Image classification with rotation-invariant variational quantum circuits. *Physical Review Research* 7:013082. <https://doi.org/10.1103/PhysRevResearch.7.013082>
- Schatzki L, Larocca M, Nguyen QT, et al (2024) Theoretical guarantees for permutation-equivariant quantum neural networks. *npj Quantum Information* 10(1):12. <https://doi.org/10.1038/s41534-024-00804-1>
- Schuurmann DJ (1987) A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics* 15(6):657–680. <https://doi.org/10.1007/BF01068419>
- Shi J, Wang W, Lou X, et al (2023) Parameterized Hamiltonian learning with quantum circuit. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45(5):6086–6095. <https://doi.org/10.1109/TPAMI.2022.3203157>
- Shin S, Teo YS, Jeong H (2024) Dequantizing quantum machine learning models using tensor networks. *Physical Review Research* 6:023218. <https://doi.org/10.1103/PhysRevResearch.6.023218>
- Sitzmann V, Martel JNP, Bergman AW, et al (2020) Implicit neural representations with periodic activation functions. In: *Advances in Neural Information Processing*

- Systems, vol 33. Curran Associates, Inc., pp 7462–7473, URL <https://proceedings.neurips.cc/paper/2020/hash/53c04118df112c13a8c34b38343b9c10-Abstract.html>
- Slabbert D, Petruccione F (2024) Hybrid quantum-classical feature extraction approach for image classification using autoencoders and quantum SVMs. arXiv preprint arXiv:2410.18814 <https://doi.org/10.48550/arXiv.2410.18814>
- Stoudenmire EM, Schwab DJ (2016) Supervised learning with tensor networks. In: Advances in Neural Information Processing Systems
- Strümpfer Y, Postels J, Yang R, et al (2022) Implicit neural representations for image compression. In: Computer Vision – ECCV 2022, Lecture Notes in Computer Science, vol 13686. Springer, Cham, pp 74–91, https://doi.org/10.1007/978-3-031-19809-0_5
- Sweke R, Recio-Armengol E, Jerbi S, et al (2025) Potential and limitations of random Fourier features for dequantizing quantum machine learning. Quantum 9:1640. <https://doi.org/10.22331/q-2025-02-20-1640>
- Tancik M, Srinivasan PP, Mildenhall B, et al (2020) Fourier features let networks learn high frequency functions in low dimensional domains. In: Advances in Neural Information Processing Systems, vol 33. Curran Associates, Inc., pp 7537–7547, URL <https://proceedings.neurips.cc/paper/2020/hash/55053683268957697aa39fba6f231c68-Abstract.html>
- Tang E (2019) A quantum-inspired classical algorithm for recommendation systems. In: Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC), pp 217–228, <https://doi.org/10.1145/3313276.3316310>
- Tilly J, Chen H, Cao S, et al (2022) The variational quantum eigensolver: a review of methods and best practices. Physics Reports 986:1–128. <https://doi.org/10.1016/j.physrep.2022.08.003>
- Ulyanov D, Vedaldi A, Lempitsky V (2020) Deep image prior. International Journal of Computer Vision 128:1867–1888. <https://doi.org/10.1007/s11263-020-01303-4>
- Wang H, Tan J, Huang Y, et al (2024) Quantum image compression with autoencoders based on parameterized quantum circuits. Quantum Information Processing 23:41. <https://doi.org/10.1007/s11128-023-04243-3>
- West MT, Heredge J, Sevier M, et al (2024) Provably trainable rotationally equivariant quantum machine learning. PRX Quantum 5(3):030320. <https://doi.org/10.1103/PRXQuantum.5.030320>
- West MT, Heredge J, Sevier M, et al (2025) Erratum: provably trainable rotationally equivariant quantum machine learning [PRX Quantum 5, 030320 (2024)]. PRX Quantum 6:020902. <https://doi.org/10.1103/g4zk-79l9>

- Xiao H, Rasul K, Vollgraf R (2017) Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. arXiv preprint arXiv:1708.07747 <https://doi.org/10.48550/arXiv.1708.07747>
- Xu R, Wang X, Chen K, et al (2021) Positional encoding as spatial inductive bias in GANs. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 13569–13578
- Yang J, Shi R, Wei D, et al (2023) MedMNIST v2—a large-scale lightweight benchmark for 2D and 3D biomedical image classification. Scientific Data 10:41. <https://doi.org/10.1038/s41597-022-01721-8>
- Yao XW, Wang H, Liao Z, et al (2017) Quantum image processing and its application to edge detection: theory and experiment. Physical Review X 7(3):031041. <https://doi.org/10.1103/PhysRevX.7.031041>
- Zhai X, Xiao T, Huang J, et al (2025) Quantum neural compressive sensing for ghost imaging. Physical Review Applied 23(1):014018. <https://doi.org/10.1103/PhysRevApplied.23.014018>
- Zhang Y, Lu K, Gao Y, et al (2013) NEQR: a novel enhanced quantum representation of digital images. Quantum Information Processing 12(8):2833–2860. <https://doi.org/10.1007/s11128-013-0567-z>
- Zhao J, Qiao W, Zhang P, et al (2024) Quantum implicit neural representations. In: Proceedings of the 41st International Conference on Machine Learning, Proceedings of Machine Learning Research, vol 235. PMLR, pp 60940–60956, URL <https://proceedings.mlr.press/v235/zhao24l.html>
- Zhou RG, Hu W, Fan P, et al (2017) Quantum realization of the bilinear interpolation method for NEQR. Scientific Reports 7(1):2511. <https://doi.org/10.1038/s41598-017-02575-6>